

Non-Autistic & Autistic children: Why should I say “*um*” if I’m not talking to anybody?

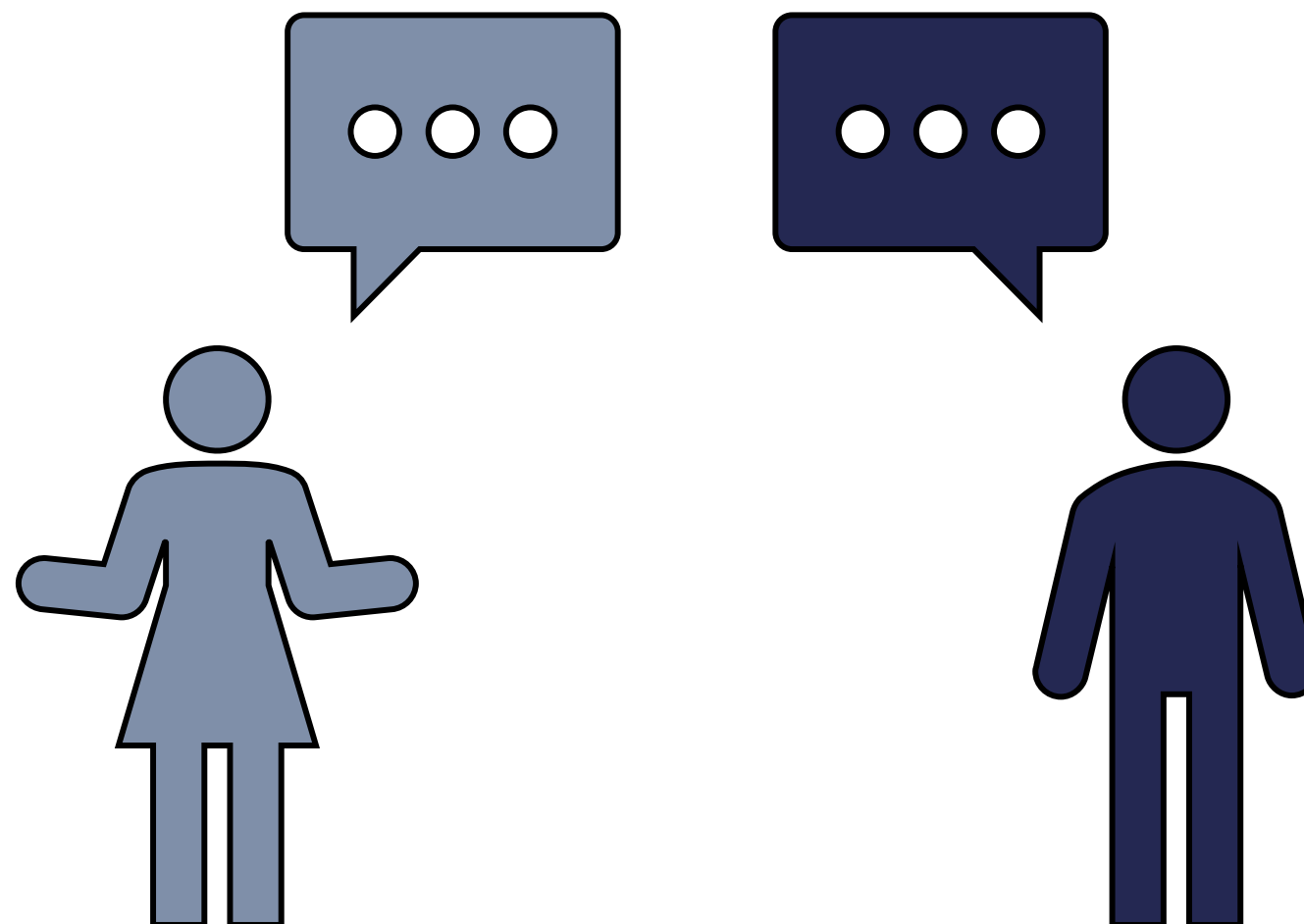
Cynthia Boo, M.A., M.S.

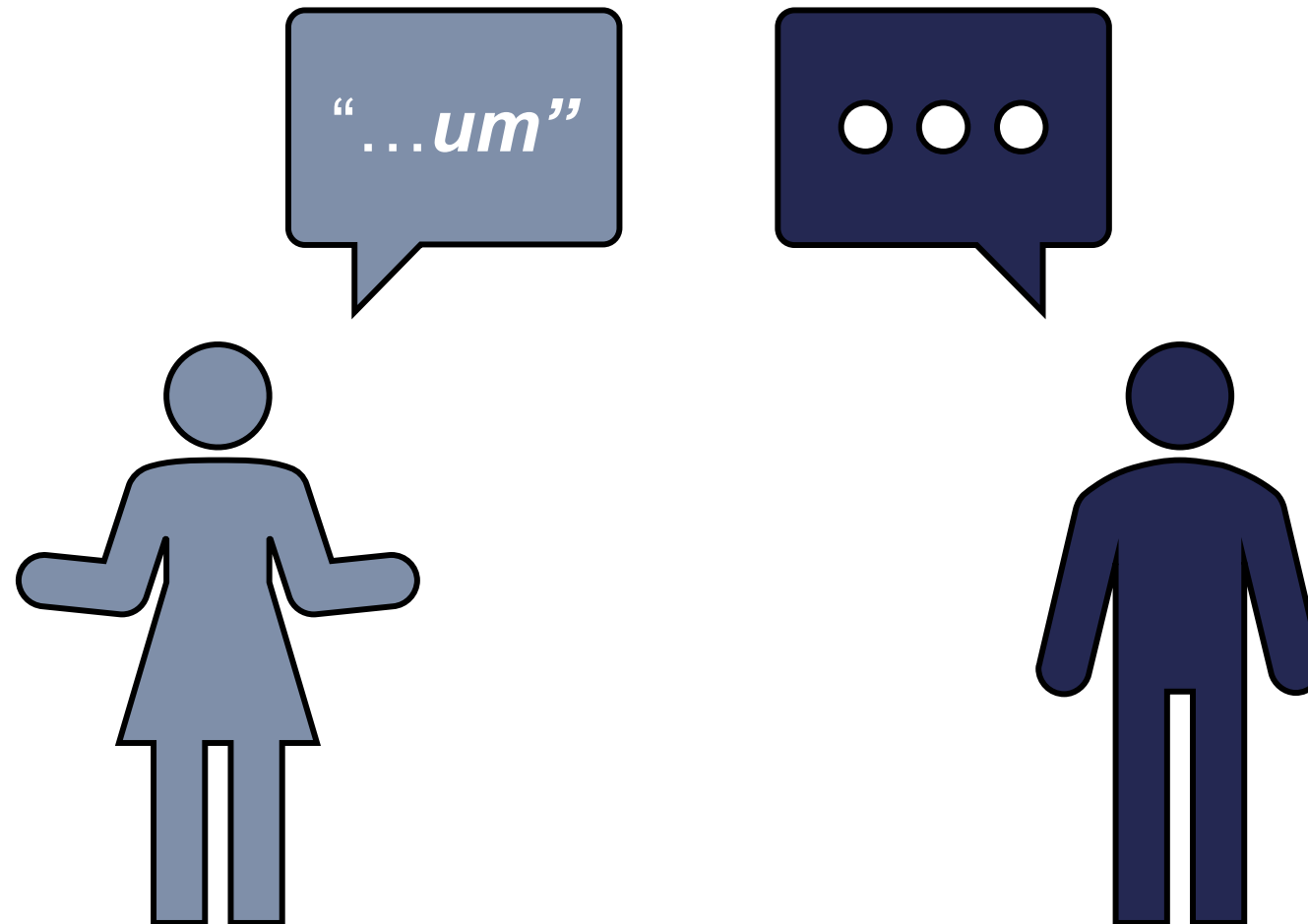
University of Connecticut – Department of Psychological Sciences

**Discourse and disfluencies in cognitive and speech disorders
18th International Pragmatics Conference**

Friday, July 14th, 2023

UConn







“...um”

What is the purpose of “*um*” in discourse?

- Communicate information about the speaker’s state, such as:
 - Conveying uncertainty (reflection of cognitive load)

My PhD advisor:	“Do you feel ready to present at IPrA?”
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Me:	“ <i>Um</i> , yeah...”
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Student in my class:	“I heard taking shrooms can make you less depressed.”
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Me:	“ <i>Um</i> , well, psychoactive mushrooms, <i>um</i> , affect the brain’s chemistry, and <i>um</i> ...why don’t we, <i>um</i> , get back to our discussion on REM sleep...”
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“...um”

What is the purpose of “*um*” in discourse?

- Communicate information about the speaker’s state, such as:
 - Conveying uncertainty (reflection of cognitive load)
- Serves as a turn management device (pragmatic device)

Turn-Taking

A: “When do you want to get lunch?”
B: “**Um**, I could do next Friday.”

Turn-Maintaining

B: “I have several dates in the next month, **um**, let me get back to you.”

Turn-Yielding

A: “I’ll eat anything, so don’t worry about me, **um...**”

Substantial body of literature looking at “*um*” use among autistic individuals

- “*Um*” use appears to be a useful linguistic tool for conversational reciprocity, which is thought to be impaired in autism
- Researchers are interested in whether “*um*” appears in autistic individuals’ speech – as a reflection of their pragmatic abilities
 - Researchers expect fewer “*um*” tokens by autistic individuals

Core characteristic of autism:

- repetitive and restricted behaviors, and
- **challenges with social communication and interaction**

American Psychiatric Association, 2013; Gorman et al., 2016; Irvine et al., 2016; Lawley et al., 2022; McGregor & Hadden, 2020 ; Tager-Flusberg et al., 2005

Research finds autistic individuals use “um” less often than non-autistic (NonAu) individuals

In discourse contexts where participants:

Irvine et al. (2016)	Gorman et al. (2016)	McGregor & Hadden (2020)
are asked to describe a painting to a research assistant	answer questions during diagnostic testing to an examiner	are asked to describe how to play their favorite sport to a research assistant

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non-interactive discourse contexts
with visual feedback (by research assistant/examiner)

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non-interactive discourse contexts
with visual feedback (by research assistant/examiner)

Past investigations have primarily focused on this one particular discourse type that is not truly conversational

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are asked to describe a painting to a research assistant	answer questions during diagnostic testing to an examiner	are asked to describe how to play their favorite sport to a research assistant

non-interactive discourse contexts
with visual feedback (by research assistant/examiner)

If “*um*” is truly a linguistic tool for conversational reciprocity, shouldn’t we investigate “*um*” use of autistic individuals in actual conversational settings?

Primary Research Question of Current Investigation:

**Can we extend the previous finding
to other discourse contexts?**

Namely, ones that:

- are **interactive** versus **noninteractive**, and
- have **visual feedback** versus **no visual feedback**

Types of discourse contexts in this investigation

	Non-Interactive	Interactive
With visual feedback	Irvine et al. (2016) Gorman et al. (2016) McGregor & Hadden (2020)	Study 2
With no visual feedback	Study 1	Study 3

****participants across all three studies were similar in terms of age and cognitive & verbal ability****

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Visual Feedback v. No Visual Feedback?

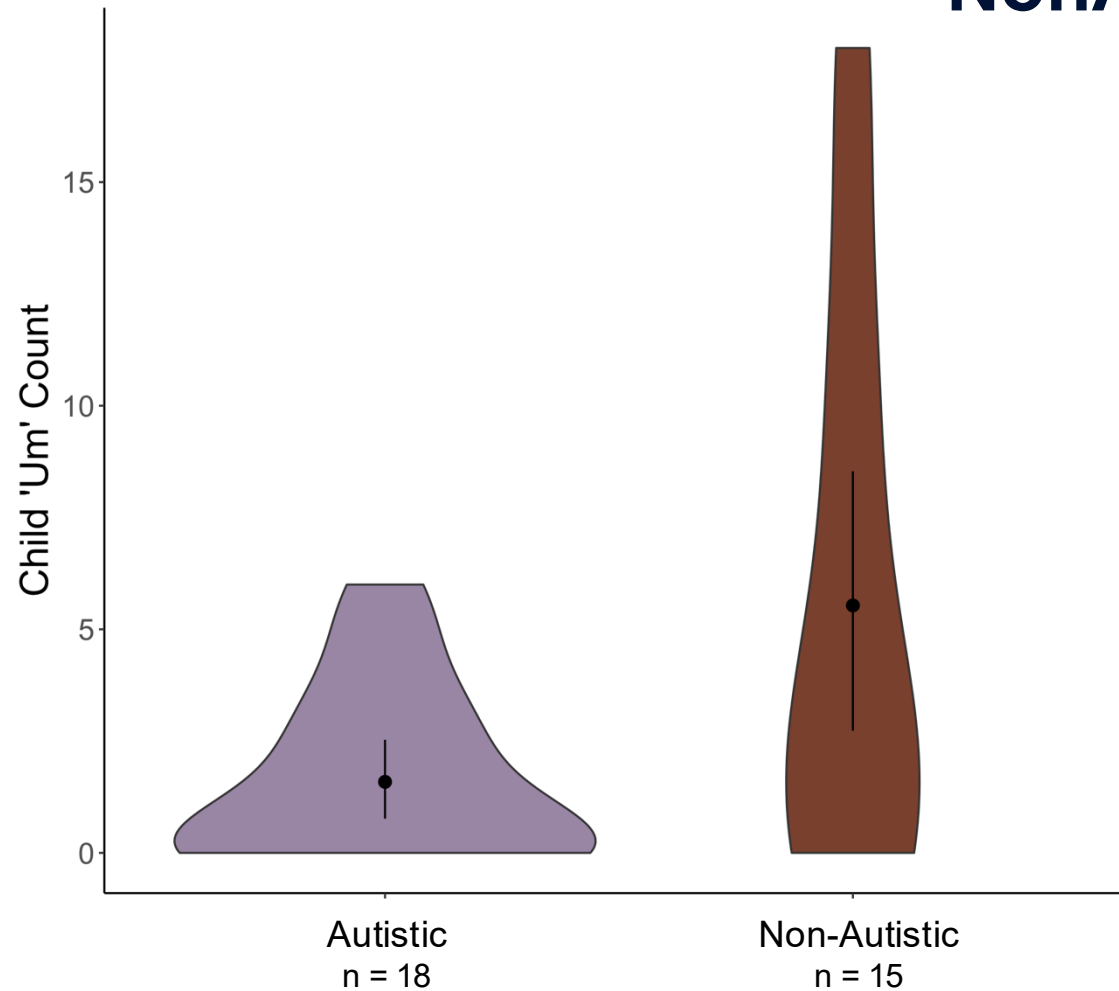
	Non-Interactive	Interactive
With visual feedback	Irvine et al. (2016) Gorman et al. (2016) McGregor & Hadden (2020)	Study 2
With no visual feedback	Study 1	Study 3

Study 1 – Non-interactive storytelling w/ no visual feedback (@ Emerson College)

- 18 Autistic teenagers ($M_{\text{age}} = 13.5$, $SD = 2.3$ years)
 - 15 NonAu teenagers ($M_{\text{age}} = 13.4$, $SD = 2.1$ years)
-
- Participants were asked to:
 - develop a fictional story, then
 - narrate story for three minutes to a panel of (prerecorded) judges who did not provide any clear audible/visual feedback (i.e., Trier Social Stress Test; Kirschbaum et al., 1993)



Autistic teenagers produced fewer “*um*” tokens than NonAu teenagers



Autism Group:
 $M = 2.00$, $SD = 2.59$

Non-Autistic Group:
 $M = 5.27$, $SD = 5.18$

$t(31) = -2.097$, $p = 0.023$, $d = -0.663$

(Note: findings remain significant even when accounting for total number of word tokens)



Study 1 – Non-interactive storytelling w/ no visual feedback

Results

Did we replicate previous findings (i.e., Autistic < Non-Autistic)?

	Non-Interactive	Interactive
With visual feedback	Irvine et al. (2016) Gorman et al. (2016) McGregor & Hadden (2020)	Study 2
With no visual feedback	Study 1 	Study 3

What about interactive v. non-interactive?

	Non-Interactive	Interactive
With visual feedback	Irvine et al. (2016) Gorman et al. (2016) McGregor & Hadden (2020)	Study 2
With no visual feedback	Study 1 	Study 3

Study 2 – Interactive conversation w/ visual feedback (@ Emerson College)

- Same participants (18 Autistic & 15 NonAu teenagers)
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- Participants answered questions about themselves

“Can you tell me what you like to do for fun?”

“Tell me about your family.”

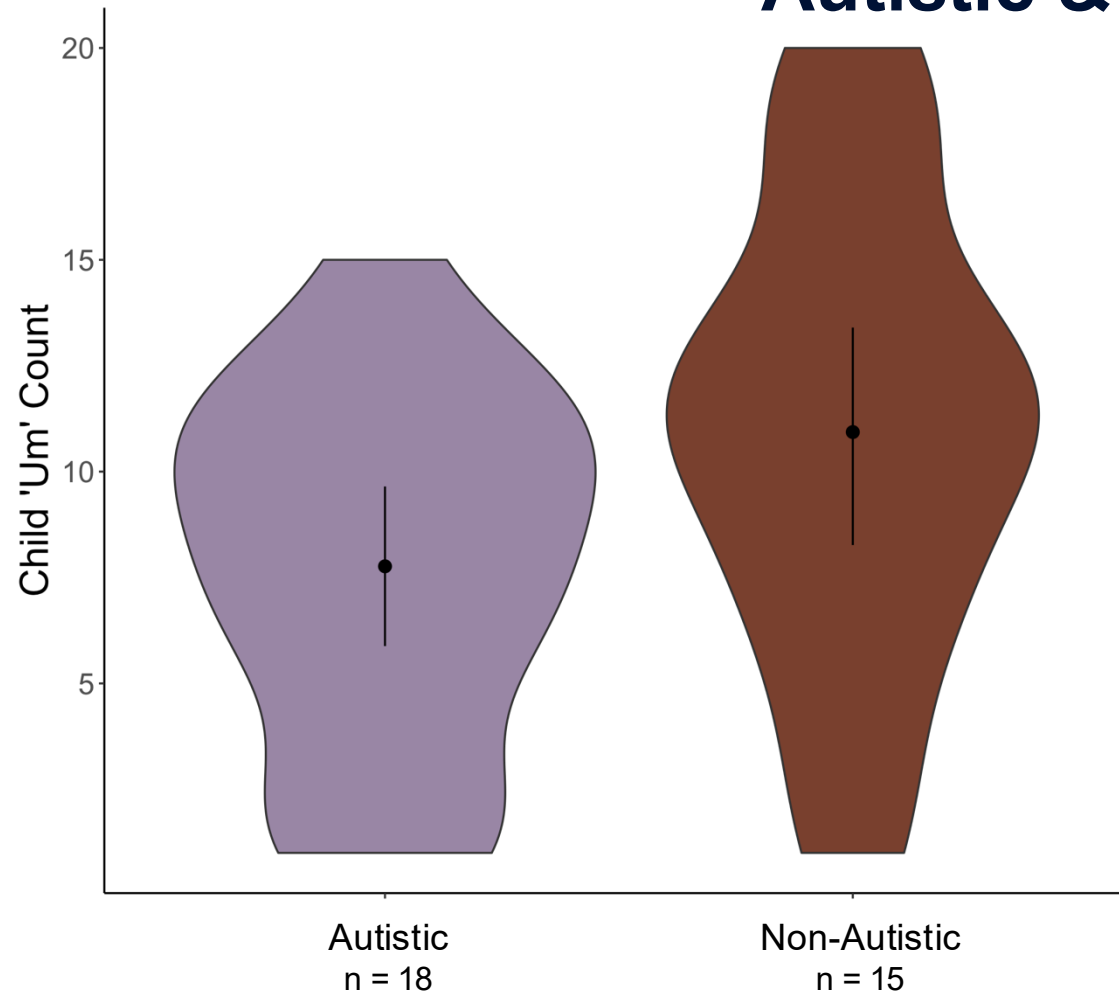
“Can you tell me about a cool trip or vacation you have been on?”

“Tell me about the hardest or your least favorite part of school.”

- Research assistants (RAs) were encouraged to respond naturally (verbally and non-verbally) to participants' answers



No significant difference in “*um*” use between Autistic & NonAu teenagers



Autism Group:
 $M = 7.76, SD = 4.22$

Non-Autistic Group:
 $M = 10.93, SD = 5.41$

$t(31) = -0.296, p = 0.769, d = 0.245$



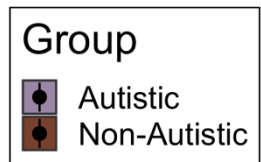
Study 2 – Interactive conversation w/ visual feedback

Cross-Task Comparison

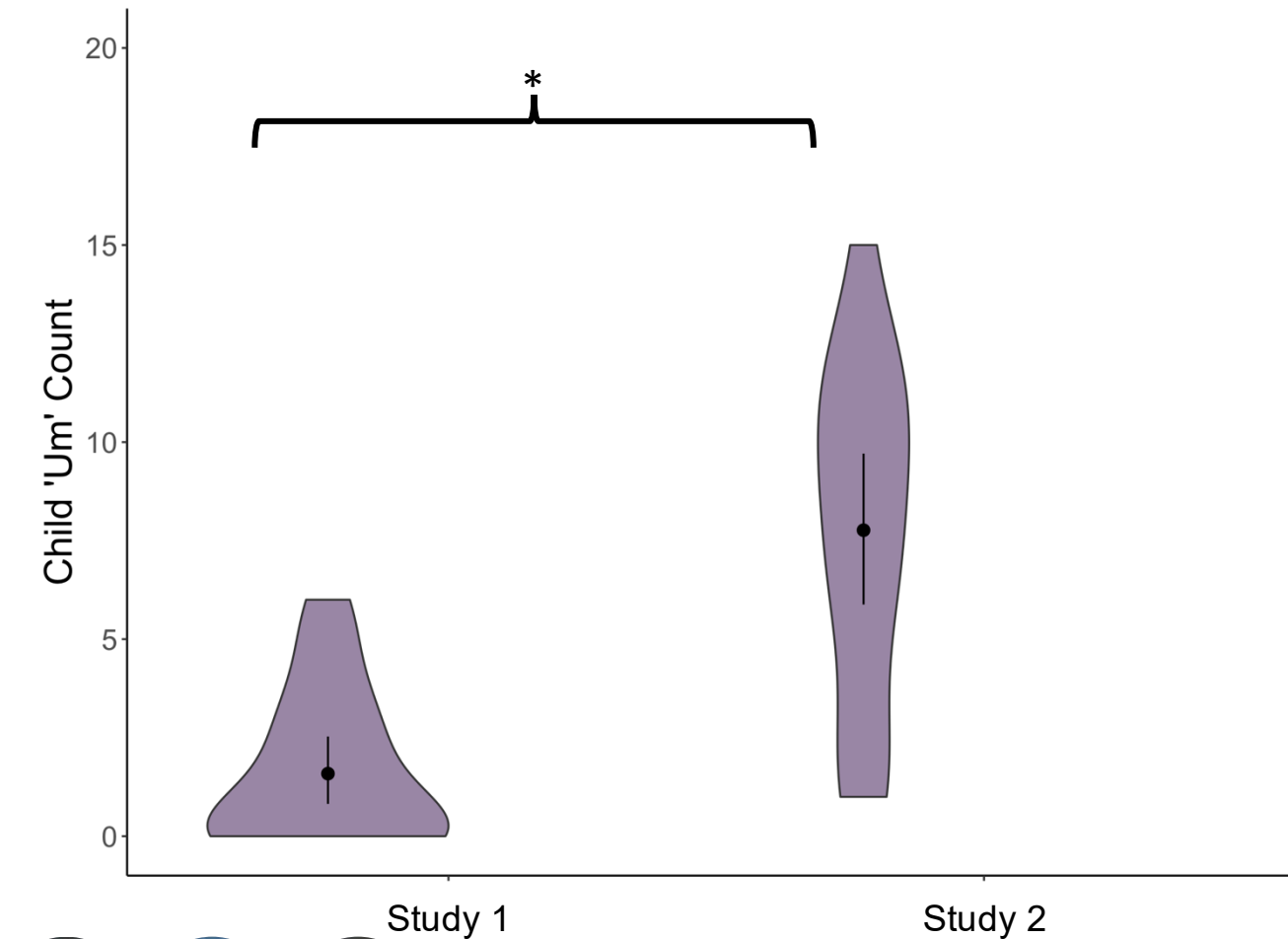
Study 1 – Non-Interactive

VS

Study 2 – Interactive



Both Autistic and NonAu teenagers produced more “*um*” tokens during the interactive than non-interactive task



Autism Group:

***Non-Interactive:* M = 2.00, SD = 2.59**

***Interactive:* M = 7.76, SD = 4.22**

Main effect:

$F(1, 28) = 31.127, p < 0.001, \eta_p^2 = 0.526$

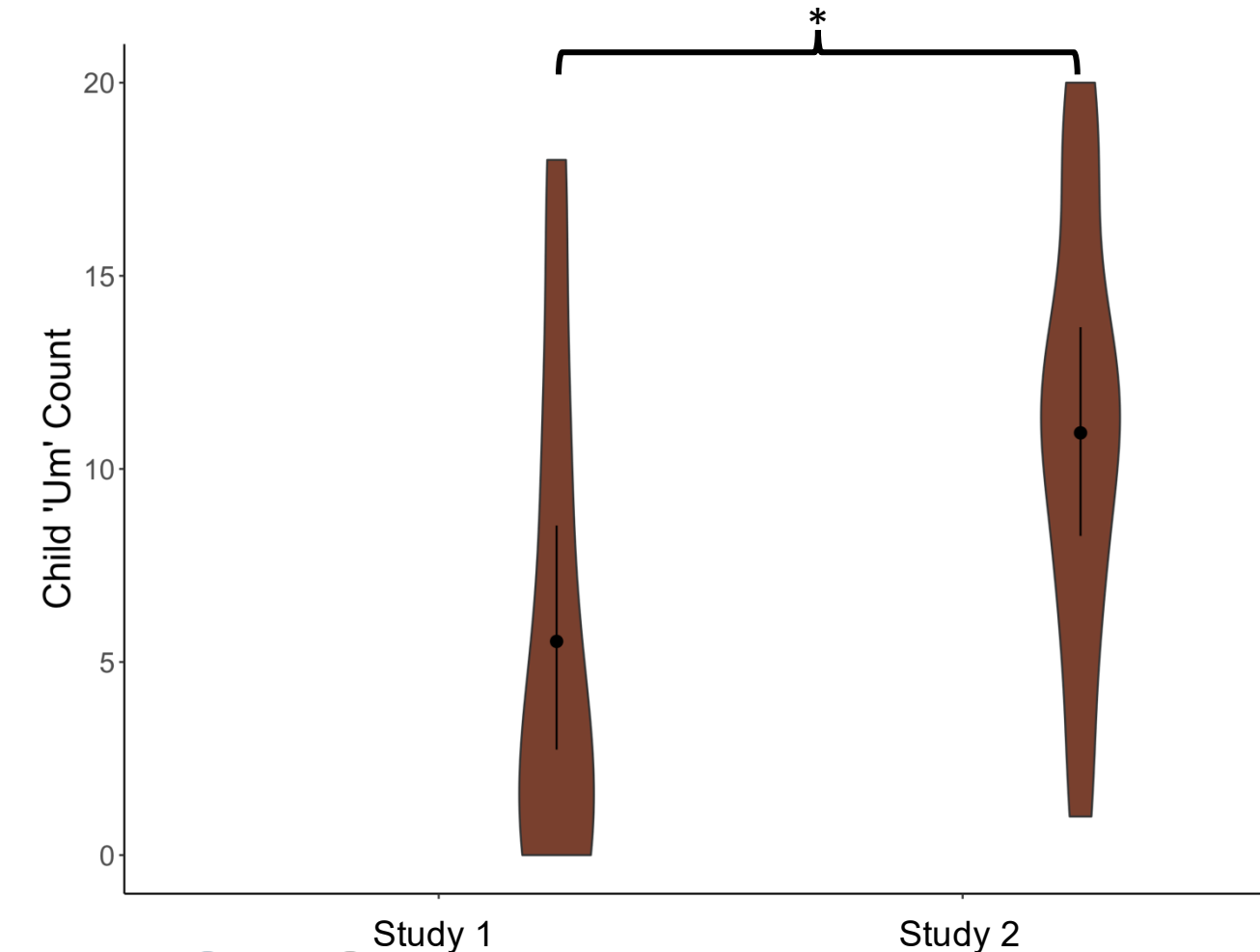
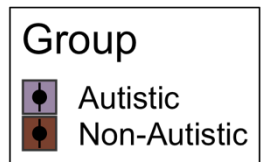
* $p < 0.05$

(Note: findings remain significant even when accounting for total number of word tokens)



Study 1 (Non-Interactive) v. Study 2 (Interactive)

Both Autistic and NonAu teenagers produced more “*um*” tokens during the interactive than non-interactive task



Autism Group:

Non-Interactive: $M = 2.00$, $SD = 2.59$

Interactive: $M = 7.76$, $SD = 4.22$

Non-Autistic Group:

***Non-Interactive*: $M = 5.27$, $SD = 5.18$**

***Interactive*: $M = 10.93$, $SD = 5.41$**

Main effect:

$F(1, 28) = 31.127$, $p < 0.001$, $\eta_p^2 = 0.526$

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Study 1 (Non-Interactive) v. Study 2 (Interactive)

Results

Did we replicate previous findings (i.e., Autistic < Non-Autistic)?

	Non-Interactive	Interactive
With visual feedback	Irvine et al. (2016) Gorman et al. (2016) McGregor & Hadden (2020)	Study 2 
With no visual feedback	Study 1 	Study 3

In fact, both groups of participants produced **MORE** “*um*” tokens in the interactive than non-interactive context

What if we remove visual feedback?

	Non-Interactive	Interactive
With visual feedback	Irvine et al. (2016) Gorman et al. (2016) McGregor & Hadden (2020)	Study 2 
With no visual feedback	Study 1 	Study 3

Study 3 – Interactive conversation w/ no visual feedback (@ UC Davis, MIND Institute)

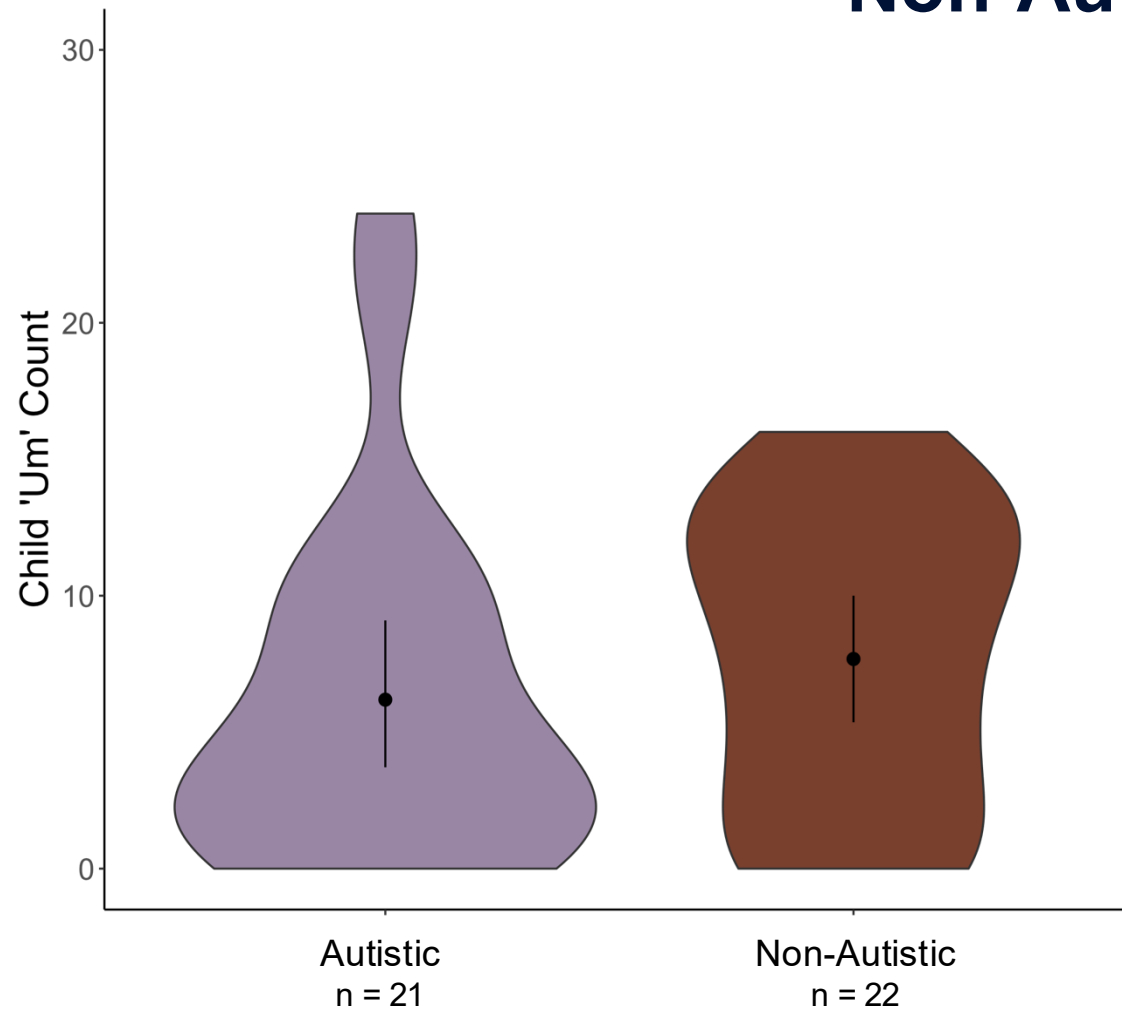
- 21 Autistic teenagers ($M_{\text{age}} = 11.6$, $SD = 2.2$ years)
 - 22 NonAu teenagers ($M_{\text{age}} = 12.5$, $SD = 2.3$ years)
-



- Virtual reality headset:
 - RAs were hidden from the participants and asked questions about their daily lives, hobbies, etc.
 - Participants were asked to direct responses to the human-like avatars in the virtual reality paradigm
 - RAs were encouraged to respond naturally verbally to participants' responses
 - Virtual reality avatars did NOT visually respond to the participants' responses



No significant difference in “*um*” use between Autistic & Non-Autistic teenagers



Autism Group:
 $M = 6.19, SD = 6.59$

Non-Autistic Group:
 $M = 7.68, SD = 5.44$

$t(31) = -0.296, p = 0.769, d = 0.247$



Study 3 – Interactive conversation w/ no visual feedback

Results

Did we replicate previous findings (i.e., Autistic < Non-Autistic)?

	Non-Interactive	Interactive
With visual feedback	Irvine et al. (2016) Gorman et al. (2016) McGregor & Hadden (2020)	Study 2 
With no visual feedback	Study 1 	Study 3 

Differences in “*um*” use did not occur in all discourse contexts

	Non-Interactive	Interactive
With visual feedback	Irvine et al. (2016) Gorman et al. (2016) McGregor & Hadden (2020)	Study 2 ✗
With no visual feedback	Study 1 ✓	Study 3 ✗

What do these findings tell us about “*um*” use in autism?
 What do these findings tell us about “*um*” use in general?

What do these findings tell us about “*um*” use in autism?

- Autistic participants use “*um*” significantly more often in interactive contexts, suggesting that they recognize **its usefulness as a turn-taking device**
 - This function of “*um*” is unnecessary in a non-interactive context, where speakers are asked to talk without interruption
- Thus, findings reveal **pragmatic strengths** for autistic speakers
 - Similar to their NonAu peers, autistic speakers use this pragmatic marker in appropriate discourse contexts, and do not use it in discourse contexts that do not afford “*um*” use

What do these findings tell us about “*um*” use in autism?

- Further, **all participants produced relatively few “*um*” tokens across all three discourse contexts** (< 1 - 3% of all word tokens), suggesting that “*um*” use may not be as useful of a marker of social communication challenges or autism as previously suggested.
- Therefore, **researchers should carefully consider the nature of the discourse context** when interpreting between-group differences of “*um*” use.

What do these findings tell us about “*um*” use, more generally?

- Findings suggest that “*um*” use differs depending on the nature of the discourse context (i.e., interactive v. non-interactive)
 - For BOTH autistic and non-autistic speakers

“*Um*” use was comparable across the two datasets with interactive tasks, suggesting that **visual feedback may not be as important as the interactive nature of the task**

**Study 2:
Interactive w/
Visual Feedback**

Autistic Group:
 $M = 7.76, SD = 4.22$

Non-Autistic Group:
 $M = 10.93, SD = 5.41$

$\approx$

**Study 3:
Interactive w/out
Visual Feedback**

Autistic Group:
 $M = 6.19, SD = 6.59$

Non-Autistic Group:
 $M = 7.68, SD = 5.44$

What do these findings tell us about “*um*” use, more generally?

- Findings suggest that “*um*” use differs depending on the nature of the discourse context (i.e., interactive v. non-interactive)
 - For BOTH autistic and non-autistic speakers
- “*Um*” use was comparable across the two datasets with interactive tasks, regardless of the presence of visual feedback
 - This suggests that verbal feedback may elicit more “*um*” use than non-verbal feedback
 - This also suggests that “*um*” may be tied more to the turn-taking mechanics of discourse

Thank you for your attention!



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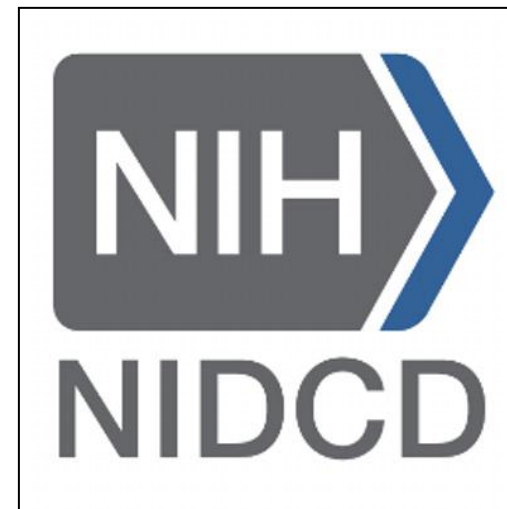
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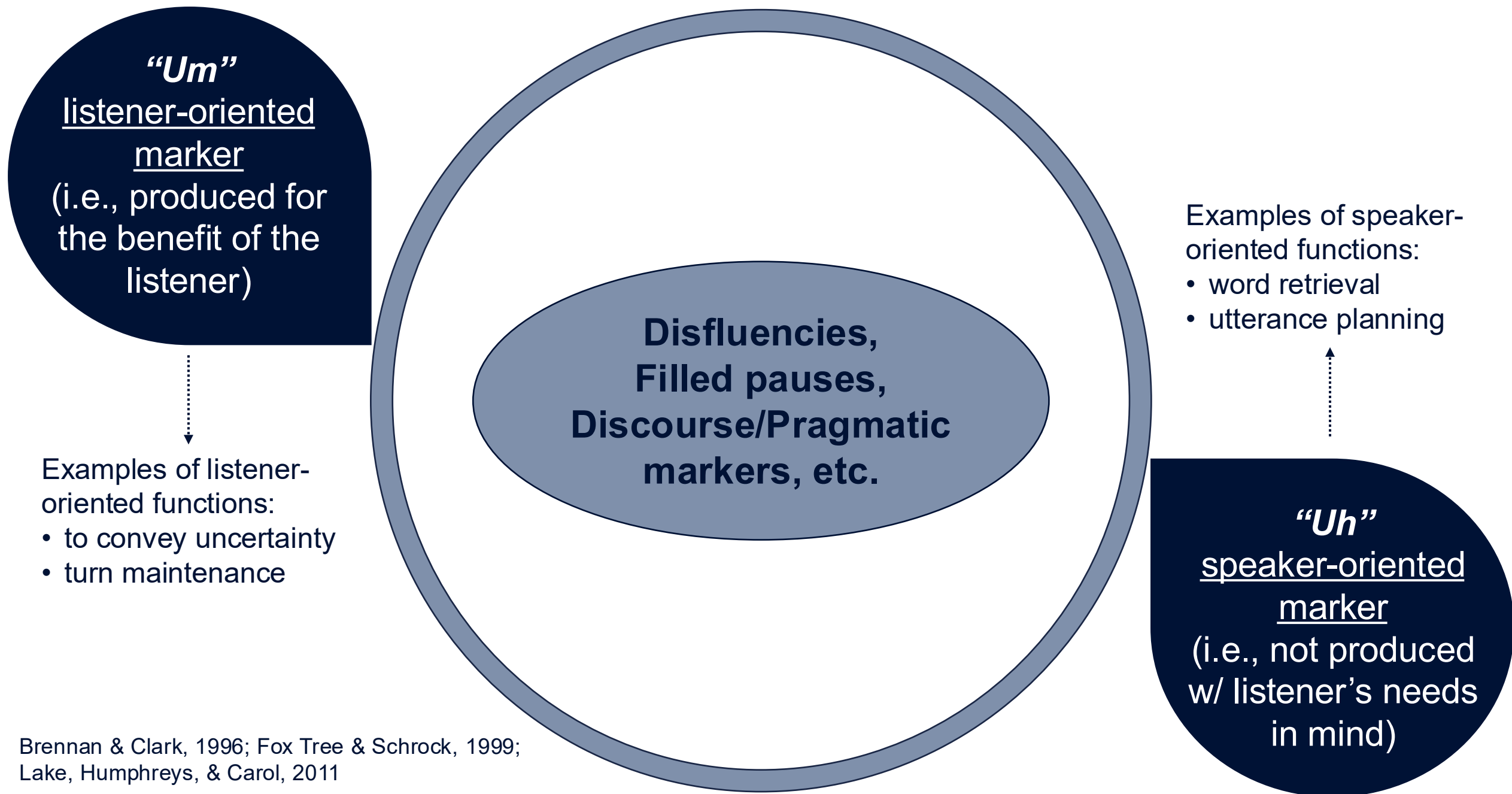


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Brennan & Clark, 1996; Fox Tree & Schrock, 1999;
Lake, Humphreys, & Carol, 2011

Means & standard deviations of average “*uh*” tokens used by participants across the three studies

	Autistic	Non-Autistic	<i>p</i>-value
Study 1 (non-interactive, no visual feedback)	2.07 (2.81)	2.60 (3.27)	0.978
Study 2 (interactive w/ visual feedback)	4.50 (4.92)	3.07 (3.77)	0.098
Study 3 (interactive w/ no visual feedback)	5.71 (6.19)	5.95 (6.15)	0.860

No significant difference in “*uh*” use between autistic and non-autistic participants in any of the studies, suggesting that “*uh*” and “*um*” may function differently in discourse.

Means (& standard deviations) of average “*um*” and “*uh*” tokens used by participants across the studies

	Autistic		Non-Autistic	
	“<i>Um</i>”	“<i>Uh</i>”	“<i>Um</i>”	“<i>Uh</i>”
Study 1 (non-interactive, no visual feedback)	2.00 (2.59)	2.07 (2.81)	5.27 (5.18)	2.60 (3.27)
Study 2 (interactive w/ visual feedback)	7.76 (4.22)	4.50 (4.92)	10.93 (5.41)	3.07 (3.77)
Study 3 (interactive w/ no visual feedback)	6.19 (6.59)	5.71 (6.19)	7.68 (5.44)	5.95 (6.15)

**significant difference between Autistic & Non-Autistic groups ($p < 0.05$)

In addition, we do not see the same task-specific differences with regards to “*uh*” use as we saw with “*um*” use.

